

Regression Algorithms for Learning & Its Application on Crypto Currency Markets

Report 1

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1. Abstract

This report presents a study of the mathematical foundations of regression algorithms with the main focus being on linear, ridge, and lasso regression. The main sources for this study were Christopher M. Bishop's *Pattern Recognition and Machine Learning* (2006) and *The Elements of Statistical Learning* by Hastie, Tibshirani, and Friedman (2008). The report explains the underlying theoretical structure and derivations of these fundamental models. The report commences with an overview of supervised learning, emphasizing the importance of training and test sets for evaluating model generalization. It then goes on to discuss the concept of cost functions and details how they guide the optimization of model parameters. A full derivation of the ridge regression formula is provided to illustrate how adding a L2

Regularization term stabilizes parameter estimates and prevents overfitting. Lasso regression is also examined for its ability to induce sparsity through an L1 penalty thus enabling automatic feature selection. The report concludes with a comparison of ordinary least squares, ridge, and lasso regression, highlighting their differences in bias–variance behavior and model complexity. Learning and understanding these concepts and principles provides the foundation for more advanced techniques in machine learning.

2. Introduction

Regression analysis forms one of the most fundamental building blocks of modern machine learning. It is used to model the relationship between one or more input variables also known as features or predictors and a continuous output variable also known as the response. The purpose of regression is to discover a function that best captures how changes in the inputs affect the output providing an opportunity for accurate predictions and interpretations.

Both Christopher Bishop (2006) and Hastie, Tibshirani, and Friedman (2008) identify regression as one of the earliest and most important methods in statistical learning. They present it not only as a predictive tool but also as a conceptual foundation for understanding more advanced algorithms.

Both texts identify generalization as a key theme. Generalization is the ability of a model to perform well on unseen data. To assess and evaluate this the datasets are split into two parts:

- Training Set: A set used to estimate the model parameters.
- Test Set: A set used to assess predictive performance on new data.

This division in the data makes sure that the model is not merely memorizing the training data but is capturing the true underlying relationship between inputs and outputs. The scenario where the model only memorizes the training data instead of understanding the underlying relationship is also known as overfitting. Good regression models achieve a balance between flexibility and simplicity, a concept described through the bias–variance tradeoff.

3. Overview of Regression Algorithms

In both Bishop and Hastie et al., regression is portrayed as a supervised learning problem, where the goal is to approximate an unknown mapping between inputs X and outputs. Mathematically, the relationship is assumed to follow:

$$Y = f(X) + \varepsilon$$

Where:

- $f(X)$ is the true function that describes the relationship between input and output variables
- ε is random noise, assumed to have mean zero and variance σ^2

The goal is to find an estimate $\hat{f}(X, \beta)$ that closely approximate $f(X)$. In this case β represents the model's parameters or coefficients.

The main difference between regression algorithms lies in how they define the model, how they measure error, and how they constrain or regularize the coefficients to prevent overfitting.

The broad overview of the process to apply regression is as follows:

1. **Model specification** which refers to deciding the form of the function, for example linear, etc.
2. **Parameter estimation** which means fitting the model using training data.
3. **Evaluation** by testing the model on new data.
4. **Regularization** which refers to adjusting the model's complexity for better generalization.

4. Linear Regression Model

Linear Regression assumes that there is a linear relationship between the input variables and the output which is being targeted. For a single observation i , the model is expressed as:

$$\hat{y}_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip}$$

where:

- $\hat{y}_i$ is the predicted output
- β_0 is the intercept (the model's bias term),
- β_j represents how strongly each input x_{ij} influences $\hat{y}_i$
- p is the number of input variables

In matrix notation this equation can be written as:

$$y = X\beta + \varepsilon$$

with:

- y being a $N \times 1$ vector of outputs received
- X being a $N \times p$ design matrix containing input features
- β being a $p \times 1$ coefficient vector
- ε being a vector of random noise

This concise and well defined representation allows efficient manipulation using linear algebra.

5. Training and Test Set

To ensure that the regression model generalizes well and does not overfit the data. We divide the data into two parts: a training set and a test set.

- The training set is used to calculate β , the vector of coefficients that minimizes the model's error.
- The test set evaluates how accurately the model predicts new, unseen data.

One of the most common measures of performance for the model is the Mean Squared Error (MSE). MSE indicates how far on average the model prediction falls when compared with the actual true values on the test set. It is calculated through the following equation:

$$\text{MSE} = \frac{1}{N_{test}} \sum_{i=1}^{N_{test}} (y_i - \hat{y}_i)^2$$

where:

- N_{test} is the total number of test data points (the size of the test set)
- y_i is the actual output value for the i^{th} test example
- $\hat{y}_i$ is the predicted value for the i^{th} test example from the regression model

A smaller MSE value would mean that the predictions by the model are closer to the values in reality indicating a well performing regression model while a larger MSE value would mean that the predictions lie further away indicating a poor performing regression model.

Low training error but high test error would indicate that the model is overfitting while high error on both sides would indicate underfitting. A systemic way of cross validation is always recommended to assess the performance of the model on multiple splits of the data.

6. Cost Functions and Ordinary Least Squares (OLS)

The cost function measures how far the model's predictions are from the true outputs. For linear regression, the cost is the sum of squared errors (SSE):

$$J(\beta) = (y - X\beta)^T (y - X\beta)$$

Where:

- y = Vector of actual (observed) outputs
- X = Design matrix or feature matrix
- β = Vector of model parameters
- $()^T$ = Transpose operator

Minimizing this function gives the line of best fit. Differentiating with respect to β :

$$\partial J(\beta) / \partial \beta = -2(X)^T (y - X\beta)$$

- ∂ = Partial Derivation Sign

Setting the derivative equal to zero would give:

$$X^T X \beta = X^T y$$

Finally solving for β gives the ordinary least squares (OLS) estimator:

$$\beta = (X^T X)^{-1} X^T y$$

This formula gives the line that minimizes the squared distance between predicted and actual values.

However, OLS can perform poorly when predictors are highly correlated, or when the dataset is high-dimensional, because the matrix $X^T X$ becomes nearly singular and unstable to invert.

This instability results in:

- Large Fluctuations in coefficient estimates with small data changes
- Overfitting where the model captures noise rather than the actual underlying pattern
- Poor generalization on test data

Regularization techniques such as Ridge and Lasso regression modify the cost function to control coefficient size and stabilize estimation.

7. Ridge Regression (L2 Regularization)

Ridge regression introduces a penalty on the squared magnitude of coefficients. The ridge cost function is:

$$J(\beta) = (y - X\beta)^T (y - X\beta) + \lambda \beta^T \beta$$

Where:

- λ = Regularization parameter

where $\lambda \geq 0$ is the regularization parameter that controls the strength of the penalty term. When $\lambda = 0$ ridge reduces to OLS while when λ is large, coefficients shrink towards zero. The L2 penalty prevents large weights thus strengthening numerical stability and reducing overfitting.

Differentiating the ridge cost function with respect to β and setting it to zero gives:

$$\partial J(\beta) / \partial \beta = -2(X)^T (y - X\beta) + 2\lambda \beta = 0$$

Rearranging terms gives:

$$(X^T X + \lambda I) \beta = X^T y$$

Therefore, when β is made the subject the result is:

$$\beta = (X^T X + \lambda I)^{-1} X^T y$$

This shows that ridge regression modifies the normal equation by adding λI , ensuring invertibility even when $X^T X$ is singular.

From a geometric viewpoint, ridge regression limits the total size of the coefficients by constraining them within a hypersphere. In probabilistic terms, Bishop (2006) shows that ridge regression corresponds to the maximum a posteriori (MAP) solution under a Gaussian prior $\beta_j \sim N(0, \tau^2)$. The regularization parameter λ is inversely proportional to this prior variance, linking the statistical and geometric interpretations.

Ridge regression slightly increases bias but greatly reduces variance, leading to better prediction on unseen data. The effective degrees of freedom in ridge regression are:

$$df(\lambda) = \text{tr}[X(X^T X + \lambda I)^{-1} X^T]$$

Where:

tr = Trace of a Matrix

I = Identity Matrix

As λ increases, $df(\lambda)$ decreases thus controlling model flexibility.

8. Lasso Regression (L1 Regularization)

Lasso Regression introduces a penalty based on the absolute values of the coefficients:

$$J(\beta) = (y - X\beta)^T (y - X\beta) + \lambda \sum_{j=1}^p |\beta_j|$$

Where:

$||$ = Absolute value

This L1 penalty encourages sparsity which means many coefficients become exactly zero, thus performing feature selection.

Geometrically, while ridge regression constrains coefficients within a circular region (L2 ball), the lasso uses a diamond-shaped region (L1 ball). The sharp corners of this shape often align with axes, forcing some coefficients to become zero. This geometric property explains why lasso yields simpler, more interpretable models.

There are no closed-form solutions for lasso regression due to the absolute value term. Instead, iterative methods are used:

- Coordinate Descent: updates each coefficient one at a time using a soft-thresholding rule.
- LARS (Least Angle Regression): computes the entire path of solutions as λ decreases.

In the special case of uncorrelated and standardized features, the solution simplifies to:

$$\begin{aligned}\beta_j &= \text{sign}(\beta_j^{\text{OLS}}) (|\beta_j^{\text{OLS}}| - \lambda), \text{ if } |\beta_j^{\text{OLS}}| > \lambda \\ \beta_j &= 0, \text{ if } |\beta_j^{\text{OLS}}| \leq \lambda\end{aligned}$$

The $\text{sign}()$ function keeps track of whether the coefficient was positive or negative before shrinkage.

It ensures lasso reduces magnitude but not direction — unless it gets shrunk all the way to zero.

Explanation:

- If the absolute value of the OLS coefficient $|\beta_j^{\text{OLS}}|$ is larger than λ , the coefficient is shrunk towards zero by the amount λ keeping its original sign.
- If it's smaller than or equal to λ then it is set to zero because anything multiplied by zero gives zero and this is the sparsity effect of lasso.

This soft- thresholding rule reduces small coefficients to zero while shrinking larger ones.

9. Comparing Regression Types, Model Evaluation & Bias–Variance Decomposition

Features	OLS	Ridge	Lasso
Penalty	None	L2: $(+\lambda \beta^T \beta)$	L1: $(+\lambda \sum_{j=1}^p \beta_j)$
Solution	Closed Form	Closed Form	Iterative
Regularization Effect	None	Shrinks Coefficients	Shrinks and sets some to zero
Bias-Variance	Low Bias High Variance	Slight Bias Reduced Variance	Sparse model Reduced Variance
Interpretability	Uses all Features	Uses all Features	Selects Key Features
When Useful	Small Data Datasets	Correlated Predictors	High Dimension Sparse Data

Evaluating regression models on unseen data is crucial. One of the recommended ways of doing this is through K-fold cross-validation, where data is divided into K folds. The model is trained on K-1 folds and tested on the remaining one and this process is repeated K times.

The average test error across folds provides a robust estimate of model performance. The value of λ that minimizes cross-validation error is chosen as the optimal regularization strength.

The expected test error can be expressed as:

$$E[(y_0 - f(x_0))^2] = \text{Bias}[f(x_0)]^2 + \text{Var}[f(x_0)] + \sigma^2$$

Bias: how far predictions deviate from the true function.

Variance(Var) : how much predictions change with different training data.

σ^2 : irreducible noise

Regularization methods like ridge and lasso keep a balance between these two sources of error hence improving generalization.

10. Conclusion

Linear, ridge, and lasso regression together form the mathematical foundation of many machine learning algorithms.

- Linear regression (OLS) fits the data directly by minimizing squared error but may suffer from high variance and overfitting.
- Ridge regression adds an L2 penalty, leading to smaller, more stable coefficient estimates and better generalization
- Lasso regression adds an L1 penalty, enforcing sparsity and automatically performing feature selection.

The derivation of the ridge estimator:

$$\beta = (X^T X + \lambda I)^{-1} X^T y$$

Shows explicitly how regularization modifies the optimization problem to produce a more robust solution.

Understanding these fundamental methods is essential before tackling advanced models such as support vector machines, kernel methods, and neural networks. These more complex algorithms all rely on the same underlying principles: fitting data through cost minimization, controlling complexity through regularization, and evaluating success through training and test performance.

11. Bibliography

1. Bishop, C. M. (2006). Pattern recognition and machine learning. Springer.
2. Hastie, T., Tibshirani, R., & Friedman, J. (2009). The elements of statistical learning: Data mining, inference, and prediction (2nd ed.). Springer.